X, Y dos v.a.

$$W = aX + bY$$

$$\mu_W = E[W] = a\mu_X + b\mu_Y \rightarrow \mu_W^2 = a^2\mu_X^2 + 2ab\,\mu_X\mu_Y + b^2\mu_Y^2$$

$$E[W^2] = E\left[(aX+bY)^2\right] = E\left[a^2X^2 + 2ab\,XY + b^2Y^2\right] =$$

$$= a^2E[X^2] + 2ab\,E[X\cdot Y] + b^2E[Y^2]$$

$$Var(W) = a^2Var(X) + b^2Var(Y) + 2ab\,Cov(X,Y)$$

$$\boxed{Var(aX+bY) = a^2 Var(X) + 2ab\, Cov(X,Y) + b^2 Var(Y)}$$

$$Var(X+Y) = Var(X) + 2\, Cov(X,Y) + Var(Y)$$

$$Var(X-Y) = Var(X) - 2\, Cov(X,Y) + Var(Y)$$

X, Y 2 v.a. con $\rho_{X,Y} = \pm 1$

$\sigma_X^2 = Var(X)$, $\sigma_Y^2 = Var(Y)$

$$Var\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = \frac{\sigma_X^2}{\sigma_X^2} - \frac{2}{\sigma_X \sigma_Y} Cov(X,Y) + \frac{\sigma_Y^2}{\sigma_Y^2}$$

$$= 1 - 2\rho_{X,Y} + 1$$

$$= 0$$

$\rho_{X,Y} = 1$

$$Var\left(\frac{X}{\sigma_x} - \frac{Y}{\sigma_y}\right) = 0 \Rightarrow \frac{X}{\sigma_x} - \frac{Y}{\sigma_y} = d \quad \text{con prob. } 1$$

$$\Updownarrow$$

$$\exists\, d \text{ tal que } P\left(\frac{X}{\sigma_x} - \frac{Y}{\sigma_y} = d\right) = 1$$

$$\frac{X}{\sigma_x} - \frac{Y}{\sigma_y} = d \therefore \boxed{Y = \frac{\sigma_y}{\sigma_x} X - d\sigma_y}$$

$$Var\left(\frac{X}{\sigma_X}+\frac{Y}{\sigma_Y}\right)=\frac{\sigma_X^2}{\sigma_X^2}+\frac{2}{\sigma_X\sigma_Y}Cov(X,Y)+\frac{\sigma_Y^2}{\sigma_Y^2}$$

$$=1+2\rho_{X,Y}+1=0$$

$$\rho_{X,Y}=-1$$

$$Var\left(\frac{X}{\sigma_X}+\frac{Y}{\sigma_Y}\right)=0 \Rightarrow \frac{X}{\sigma_X}+\frac{Y}{\sigma_Y}=d$$ con prob. 1.

$$\therefore \boxed{Y=-\frac{\sigma_Y}{\sigma_X}X+\sigma_Y d}$$ c.q.d.

$$\boxed{\rho_{X,Y} = \pm 1 \Rightarrow \exists\, d \in \mathbb{R} \text{ tal que } Y = \rho_{X,Y}\,\sigma_Y \left[\frac{X}{\sigma_X} - d\right] \text{ con prob. } 1}$$

$$\rho_{X,Y} = \pm 1 \Rightarrow Y = aX + b \text{ con prob. } 1, \; a \neq 0$$